21-Day React + TypeScript Roadmap

Day 1

TypeScript Basics

- What is TypeScript?
- Why TypeScript?
- Type Inference
- string
- number
- boolean
- null
- undefined

Practice:

- Variables
- Simple Functions

Project:

- Calculator Logic (No UI)

Day 2

TypeScript Objects & Functions

- Objects
- Arrays
- Function Types
- Optional Parameters
- Default Parameters

Practice:

- Student Object
- Product Object

Project:

- Student Management System

Day 3

Interfaces & Type Aliases

- interface
- type
- Union Types
- Literal Types

Practice:

- User Model
- Product Model

Project:

- User Profile Management

Day 4

React + TypeScript Setup

- Vite + React + TS
- Typing Components
- Typing Props
- Typing Children

Project:

- Profile Card App

Day 5

Forms & Events

- useState Types
- Input Types
- Event Types
- Controlled Components

Project:

- Registration Form

Day 6

useEffect & API Calls

- useEffect
- Dependency Array
- Cleanup Function
- Fetch API

Project:

- Weather App (TypeScript)

Day 7

useRef + useMemo + useCallback

- useRef
- useMemo
- useCallback
- React.memo

Project:

- Search Filter App

Week 2

Day 8

Custom Hooks

- Creating Hooks
- Reusable Logic

Project:

- Currency Converter (TS)

Day 9

Context API

- createContext
- useContext
- Provider Pattern

Project:

- Context Todo App

Day 10

React Router

- BrowserRouter
- Routes
- Route
- Link
- NavLink

Project:

- Multi Page Blog UI

Day 11

Advanced TypeScript

- Generics
- Generic Functions
- Generic Components

Project:

- Reusable Table Component

Day 12

Utility Types

- Partial

- Pick
- Omit
- Record
- Readonly

Project:

- User Dashboard

Day 13

Axios

- API Layer
- Service Layer
- Error Handling

Project:

- GitHub Finder App

Day 14

React Query

- useQuery
- useMutation
- Query Keys
- Caching

Project:

- Weather App V2

Week 3

Day 15

Redux Toolkit Basics

- Store
- Slice
- Actions

- Reducers

Project:

- Redux Todo App

Day 16

Redux Toolkit + TypeScript

- Typed Store
- RootState
- AppDispatch

Project:

- Expense Tracker

Day 17

RTK Query

- API Slice
- Query
- Mutation

Project:

- Blog CRUD Dashboard

Day 18

React Hook Form

- useForm
- Register
- Validation

Project:

- Signup Form

Day 19

Zod

- Schema Validation
- Error Handling

Project:

- Login + Signup System

Day 20

Performance Optimization

- React.memo
- useMemo
- useCallback
- Lazy Loading
- Suspense

Project:

- Product Catalog

Day 21

Architecture & Production Patterns

- Folder Structure
- Feature Based Design
- Environment Variables
- Reusable Components
- Service Layer
- Custom Hook Architecture

Project:

- Refactor Old Projects

Major Project 1 (Days 22-25)

Expense Tracker Pro

Features:

- Authentication
- Redux Toolkit
- RTK Query
- Charts
- Filters
- Categories
- Reports

Major Project 2 (Days 26-30)

Blog Platform

Features:

- Authentication
- CRUD Blogs
- Comments
- Likes
- User Profiles
- Admin Dashboard

After Day 30

Move To:

1. Next.js
2. Authentication (JWT)
3. Testing (Jest + RTL)
4. Docker Basics
5. CI/CD
6. System Design
7. Full Stack Projects